Own Cloud Storage on Laptop

What Is Used

a. Laptop: Windows 11 with 512 GB SSD
b. Software: Docker Desktop, Nextcloud Server 27, Ngrok, MariaDB 10.6
c. Script: PowerShell automation script using WebDAV

Step 1: Installing Docker

First, I went to docker.com and downloaded docker desktop for my windows system.

After installing it, I verified that engine was running properly. This software allows me to run nextcloud inside a container without disturbing my main windows system. It keeps everything isolated and clean.

I created a folder in my C drive:

C:\nextcloud_docker

This is the main folder where all configuration and cloud data will stay

Step 2: The Configuration

Inside that folder, I created a file named docker-compose.yml. This file tells docker how to create and run the nextcloud server and database.

docker-compose.yml :

```
services:
  db:
    image: mariadb:10.6
    container_name: nc_db
    restart: always
    command: --transaction-isolation=READ-COMMITTED
    environment:
      MYSQL_ROOT_PASSWORD: rootpass
      MYSQL_DATABASE: nextcloud
      MYSQL_USER: ncuser
      MYSQL_PASSWORD: ncpass
    volumes:
      - ./db:/var/lib/mysql
  app:
    image: nextcloud:27
    container_name: nc_app
    restart: always
    ports:
      - "8081:80"
```

```
    environment:
      MYSQL_HOST: db
      MYSQL_DATABASE: nextcloud
      MYSQL_USER: ncuser
      MYSQL_PASSWORD: ncpass
    volumes:
      - ./data:/var/www/html
      - C:/cloud_storage:/cloud_storage
    depends_on:
      - db
```

Explanation of Configuration

1. Database : uses MariaDB 10.6
2. Application: nextcloud 27 server
3. Port mapping: 8081:80 - access via port 8081 on windows,

Volumes

1. ./db:/var/lib/mysql --> database files stored in C:\nextcloud_docker\db
2. ./data:/var/www/html --> nextcloud data stored in C:\nextcloud_docker\data
3. C:/cloud_storage:/cloud_storage --> external storage mount point

Starting Nextcloud

1. Powershell command:

   cd C:\nextcloud_docker

   docker compose up -d

2. go to browser : http://localhost:8081

It asked me to create an admin account:

1. username: admin

2. password:*******

3. data folder: ‘empty’

4. database: MySQL/MariaDB

5. database user: ncuser

6. database password:ncpass

7. database name: nextcloud

8. database host:db

Then clicked Instal. nextcloud server was successfully running.

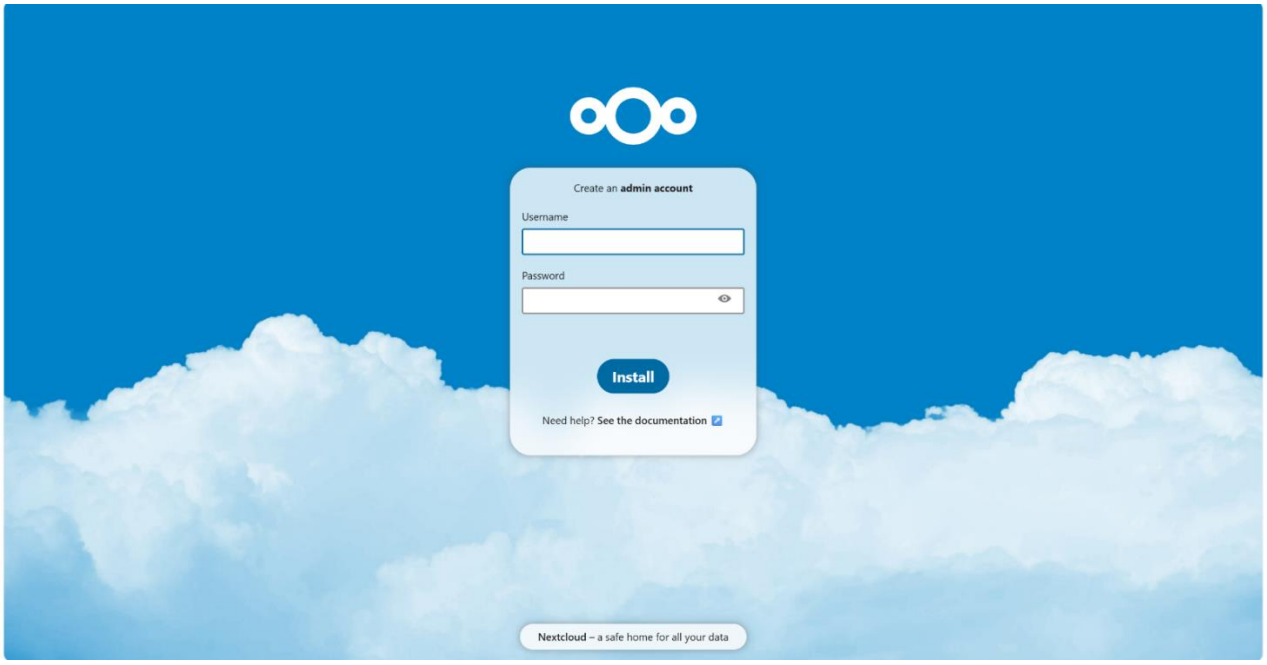


Fig:Nextcloud dashboard

Step 2.1: Configuring the "cloud inbox"

In my docker-compose.yml, I mapped a folder (`C:/cloud_storage`), but nextcloud does not show it by default. I had to manually connect it inside the nextcloud settings.

Steps:

1. Logged into nextcloud as admin

2. Navigated to apps and enabled External storage support

3. Administration settings → External Storage

4. Configured a new "Local" storage:

   1. folder name: cloud_inbox

   2.configuration: /cloud_storage

   3.authentication: none

   4.available for:admin

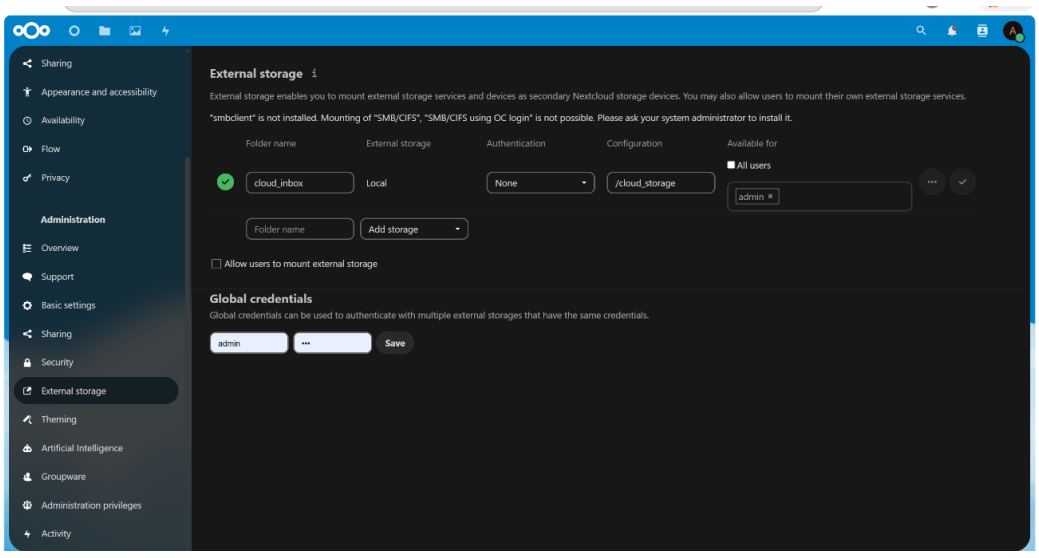


Fig:config setting

Step 2.2: Multi-user access (Isolated)

To verify the multi-user capability, I created a separate account.

Steps:

1. logged in as admin

2. navigated to admin setting --->users

3. added new end-user named ;user1

Nextcloud automatically created a dedicated folder.

Each user can only access their own files - complete isolation is maintained

Path: C:\nextcloud_docker\data\data\user1\files

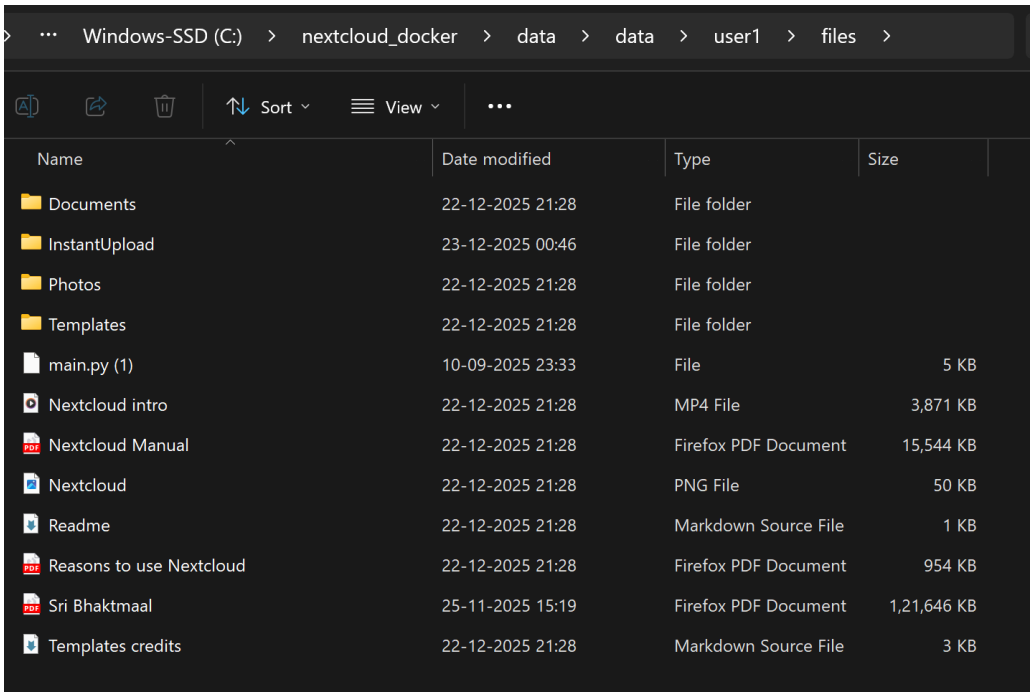


Fig: user1's dedicated space

Step3 :Getting Internet Access

My laptop is connected to wifi, so the ip address changes sometimes. Also, I wanted to access my cloud from outside my home network (WAN).

Using Ngrok for External Access

1. download ngrok from https://ngrok.com/
2. In cmd: ngrok http 8081

   Result : https://unexpectorated-letisha-geodetically.ngrok-free.dev

   Important otes:

   1. free ngrok sessions expire after 2 hours

   2. url changes every time you restart ngrok

2. Modified the trusted_domains section to include my current ip and the ngrok url:

```
'trusted_domains' =>array (
    0 => 'localhost:8081',
    1 => '127.0.0.1',
    2 => '10.186.74.67',
    3 => 'unexpectorated-letisha-geodetically.ngrok-free.dev',
),
```

3. Also added this line to fix the redirect loop

```
'overwrite.cli.url' => 'https://unexpectorated-letisha-geodetically.ngrok-free.dev',
```

4. After retart

Result: Now my cloud was accessible from anywhere using the internet.

Step4 : Automation script (Admin Only - Windows)

I wanted files in the cloud_inbox folder to automatically move to the main storage folder. So I wrote a powershell script using WebDAV.

File location: C:\nextcloud_docker\scripts\nextcloud_ingest.ps1

```
$user = "admin"
$pass = "***********"
$pair = "${user}:${pass}"
$bytes = [System.Text.Encoding]::ASCII.GetBytes($pair)
$base64 = [Convert]::ToBase64String($bytes)
$authHeader = "Basic $base64"
$sourceBase = "http://localhost:8081/remote.php/dav/files/admin/cloud_inbox/"
$destBase   = "http://localhost:8081/remote.php/dav/files/admin/"
Write-Host "Listing files from cloud_inbox..."
$req = [System.Net.HttpWebRequest]::Create($sourceBase)
$req.Method = "PROPFIND"
$req.Headers.Add("Authorization", $authHeader)
$req.Headers.Add("Depth", "1")
$res = $req.GetResponse()
$reader = New-Object System.IO.StreamReader($res.GetResponseStream())
$xml = $reader.ReadToEnd()
$res.Close()
```

```
$matches = [regex]::Matches($xml, "<d:href>(.*?)</d:href>")
$files = @()
foreach ($m in $matches) {
    $href = $m.Groups[1].Value
    if ($href -match "/cloud_inbox/" -and $href -notmatch "/$") {
        $name = ($href -split "/")[-1]
        $files += [uri]::UnescapeDataString($name)
    }
}
foreach ($file in $files) {
    Write-Host "Processing $file"
    $copyReq = [System.Net.HttpWebRequest]::Create("$sourceBase$file")
    $copyReq.Method = "COPY"
    $copyReq.Headers.Add("Authorization", $authHeader)
    $copyReq.Headers.Add("Destination", "$destBase$file")
    $copyReq.GetResponse().Close()
    $delReq = [System.Net.HttpWebRequest]::Create("$sourceBase$file")
    $delReq.Method = "DELETE"
    $delReq.Headers.Add("Authorization", $authHeader)
    $delReq.GetResponse().Close()
}
Write-Host "Ingest complete!"
```

Logic :

1.Use WebDAV PROPFIND to list files in inbox

2. Copy files to main folder

3. Delete original file from inbox

4. Authentication is done using app password generated inside nextcloud

App password:

1. log into nextcloud as admin

2. enter setting

3. navigate to security

4. go to devices and sessions

5. app name: powershell automation

6. click create new app password

Setting Task scheduler:

1. Task Scheduler
2. Enter create task
3. Enter name
4. High privilege
5. Set time
6. Start a programm
7. Argument: -ExecutionPolicy Bypass -WindowStyle Hidden -File "C:\nextcloud_docker\scripts\nextcloud_ingest.ps1"
8. Done

Step5 : Mobile Access

One way:

1. LAN: The browser access using : http://10.186.74.67:8081
2. WAN: using : https://unexpectorated-letisha-geodetically.ngrok-free.dev

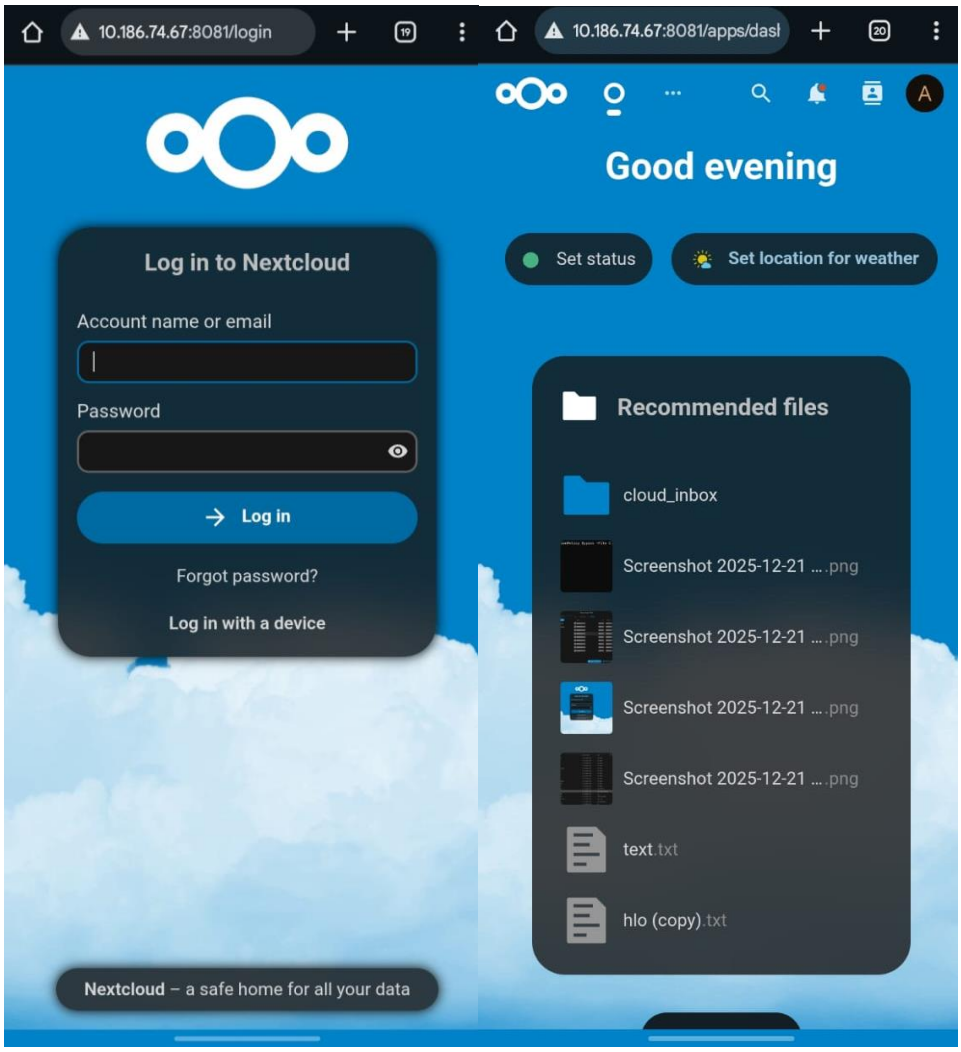


Fig:Mobile access

Another way:

1. Download nextcloud from play store
2. Choose login and enter : https://unexpectorated-letisha-geodetically.ngrok-free.dev
3. Fill credentials
4. Done
5. For-auto upload :

a) Select enable auto upload
b) Select folder(I choosed screenshot)
c) Unmetered wifi

The data resides in : C:\nextcloud_docker\data\data\user1\files\InstantUpload\Screenshots

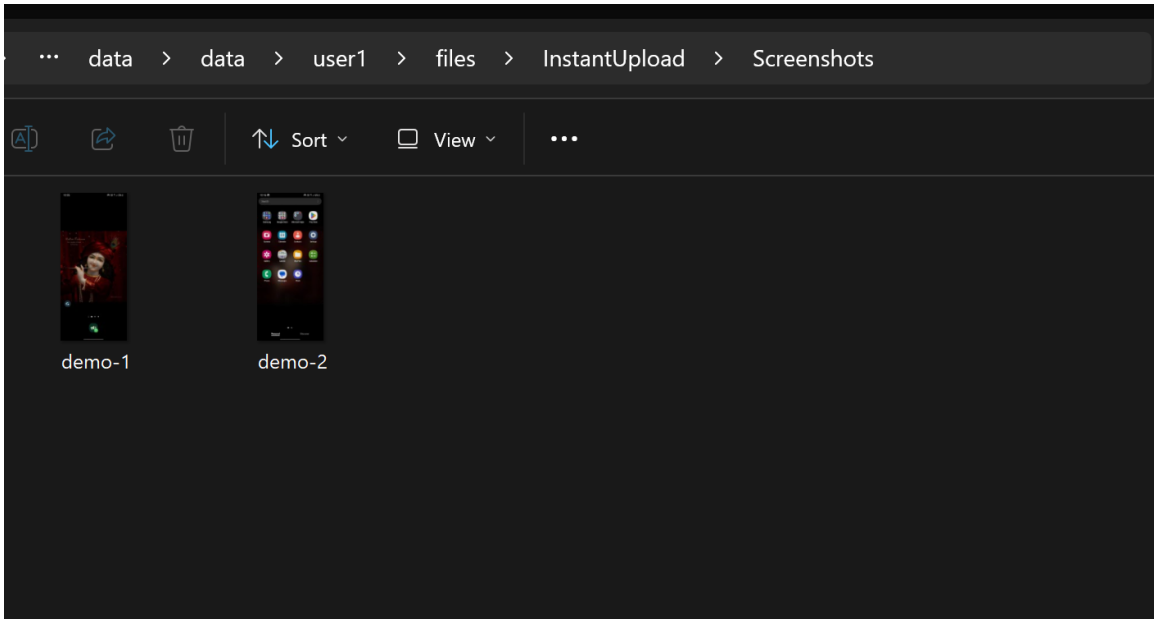


Fig: screenshot folder for user1(the another end user except admin)

Every user get isolated with their own space as proof in the screenshot, where admin and user1 have individual storage:

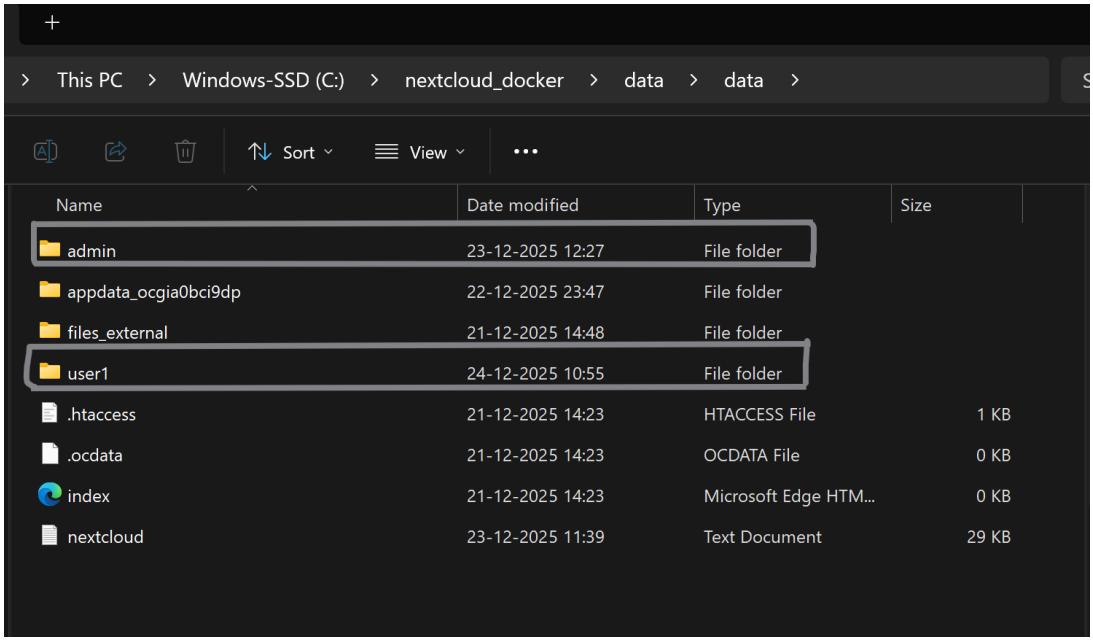


Fig: two different folders for different end users (admin and user1)

Conclusion:

I basically built my own free version of Google Drive. Now my phone photos automatically save straight to my PC's SSD. It feels good knowing that 100% of my data stays with me, giving me total privacy without depending on any big company servers.